

MACHINE TOOLS LAB

III B. Tech. II Semester: MECHANICAL ENGINEERING

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6ME34	PCC	0	0	2	1	40	60	100

COURSE OVERVIEW:

This course is introduced to create awareness on various machining operations, equipment and their working to create prototype models.

COURSE OUTCOMES:

At the end of the course, the student should be able to:

1. To identify various cutting tools and its uses.
2. To understand the working of various machines such as Lathe, Shaper, Slotter and Planing.
3. To understand the working of various operations on Milling and Drilling Machines
4. To make students to create prototype models by CNC machining.

LIST OF EXPERIMENTS:

1. Step turning and taper turning using lathe machine.
2. Thread cutting and knurling using lathe machine.
3. Drilling and step boring using lathe machine.
4. Drilling and Tapping using drilling machine.
5. Shaping of V groove using shaper.
6. Slotting of a keyway using slotter machine.
7. Milling of gear.
8. Surface Grinding.
9. Planing Machine operations
10. Cylindrical grinding Operations.
11. Tool and cutter grinding operations
12. To perform different operations by using CNC lathe machine
13. To perform different operations by using CNC Milling machine